

# NLP: A Quick Look at Term Frequency-Inverse Document Frequency

Natural language processing (NLP) helps computers understand and generate written and spoken words. NLP uses Term Frequency-Inverse Document Frequency to analyse the importance of a word within a document by assigning a numerical score to each word. Let's find out how...



**T**erm Frequency-Inverse Document Frequency (TF-IDF) is a method to convert text into numerical features. TF-IDF assigns higher weightages to important words in a corpus and reduces the importance of general words like 'the', 'a', 'if' by assigning lesser importance to them.

To establish the importance of a word, this approach first determines 'how often the word appears in a document'; then it searches all the documents to determine 'how rare the word is across all documents'. The first exercise is known as Term Frequency (TF) and the second is known as Inverse Document Frequency (IDF). The multiplication of these two parameters, i.e.,  $TF \times IDF$ , provides the weightage of that word. If a word is frequent in a specific document but rare across others, then the word is more important and gets a higher weightage.

This strategy helps us to remove the bias from frequently used uninformative words (e.g., 'the', 'is', 'are'). It also identifies keywords of a document. This helps us to classify a document to identify its topic model by retrieving meaningful keywords from the target documents.

## Simple example of TD-IDF

Assume we have three documents:

Doc1: "this magazine is great"

Doc2: "this magazine is bad"

Doc3: "The quality of this magazine is great"

### Step 1: Vocabulary

Unique terms = {this, magazine, is, great, bad, quality, of}

| Term     | Doc1 | Doc2 | Doc3 |
|----------|------|------|------|
| this     | low  | low  | low  |
| magazine | high | high | high |
| is       | low  | low  | low  |
| great    | high | 0    | high |
| bad      | 0    | high | 0    |
| quality  | 0    | 0    | high |
| of       | 0    | 0    | high |